

James Schaffer

jameesschaffer.dev@gmail.com

Personal Profile

Second year student at University of Leicester studying Computer Science. Am enthusiastic, industrious and have worked in a team and independently. Interested in many aspects of computer science such as digital images and video game development, taking the initiative to teach myself 2 engines and 3 programming languages to develop games. Completed 4 games now, 3 of which I submitted to competitions, with Escape the Darkness being nominated for BAFTA YGD's Game making award in 2023.

Projects and other experience

Portfolio Website 2023-2025

- Created a personal portfolio website to showcase projects, first in 2023 after BAFTA YGD nomination, using HTML, CSS & JS and hosted using google cloud with cloudflare
- Re-developed in 2025 using HTML, CSS & JS and hosted on github pages with cloudflare
- Using : HTML, CSS, JS, Github pages, Google cloud (old site), Cloudflare
- Website link : www.jameesschaffer.dev
- Github link : github.com/James-Schaffer/James-Schaffer.github.io

Random distribution & Cantor paring research project 2025

- Project researching, developing and benchmarking different random point distribution algorithms and mapping infinite 2D arrays to 1D arrays (cantor paring). Built benchmarking tools in python to test the different algorithms
- Using : Python, Matplotlib
- Link : jameesschaffer.dev/html/portfolio/programming/mappingInfinite2DArraysToA1DArray.html

Hackathons (IBM 2025, Capital One 2024, Morgan Stanley 2024)

- Took part in multiple team based programming challenges, we were required to complete a task/develop software to meet requirements in a short amount of time, usually around 2 days within a randomly assigned group
- Came up with solutions and helped organise the team in order as well as work effectively in a group
- **2nd in Morgan Stanley coding challenge and won rookie innovator in IBM hackathon**
- Skills: problem solving, teamwork, collaboration, leadership, planning, organisation, communication

Recipe timer application 2024 - A-level coursework

- Developed a mobile application and wrote a full report of the project involving the analysis, research, design and development processes
- Provided the user with a recipe sharing platform with built in cooking support in the form of an responsive timer adapting to how long you take on each part of the recipe
- Implemented an account system which allowed users to personalise their experience and find new recipes
- Using : Python, Kivy, SQLite
- Link to project and reflection : jameesschaffer.dev/html/portfolio/programming/recipeManagerApp.html

Escape the darkness 2023

- Developed as short 2D puzzle-survival game, nominated for **BAFTA YGD 2023** 15-18 game making category, featured on local news
 - Most code and all art assets created by me. Built many core systems including a real-time FOV mask which enabled many of the main gameplay mechanics to function
 - Using : Unity, C#, Aseprite
 - Link : jameesschaffer.dev/html/portfolio/games/escapeTheDarkness.html
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Education

University of Leicester : September 2024-Present

BSc Computer Science with year in Industry

Completed 1st year with an overall mark of 80% (**equivalent to a 1st**)

- Year 1 modules : Computer Fundamentals , Programming Fundamentals , Mathematical Fundamentals , Computer Architecture , Introduction to Object Oriented Programming , Requirements Engineering and Professional practice , Algorithms, Data Structures and Advanced Programming , Foundations of Computation

Acting course rep for second year computer science : September 2025 - present

- Acting as a voice and representative for year 2 computer science students
 - Gathering feedback from students and passing it to appropriate people/groups to incite positive change and pass back information to the students
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Achievements

- **Rookie innovator award** and **finalist** at **IBM** x Leicester CS society hackathon 2025
 - **Morgan Stanley** Leicester Coding Challenge 2024 **2nd place**
 - **Gold DofE** - 7/9/2019 - 25/03/2024
 - 2023 **BAFTA YGD Finalist** - Escape The Darkness
 - Bronze DofE - 7/9/2019 - 9/3/2022
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Licences & certifications

- IBM MQ Developer essentials - 2025
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Interests

- Computer game development (4 complete games)
- Computer programming (many projects in areas such as : procedural generation, life simulations, web development)
- Member of Computer Science Society in Leicester